
Condition-Based Integration of In Vivo RNA Delivery Data Enables AI-Guided Ionizable Lipid Design



Abstract

Lipid nanoparticles (LNPs) are widely used for RNA-based therapeutics, where ionizable lipids (ILs) play a critical role in enabling efficient cytoplasmic RNA delivery through pH-dependent mechanisms. However, exhaustive in vivo screening of ionizable lipid candidates is costly and time-consuming, motivating the development of AI-based models for predicting RNA transfection potency from lipid structure. A major challenge in this setting is the limited availability of in vivo data and the substantial heterogeneity across experimental conditions, which complicates direct aggregation of datasets in the absence of shared reference LNPs. In this work, we investigate condition-based integration of heterogeneous in vivo datasets and demonstrate that datasets sharing the same cargo type, delivered gene, target organ, and administration route can be meaningfully pooled despite differences in LNP composition. Leveraging this strategy, we train AI models that achieve strong within-condition predictive performance, enabling reliable ranking and prioritization of ionizable lipid candidates. Our results provide a practical framework for expanding training data for LNP design under data-limited in vivo settings and highlight both the potential and limitations of AI-driven prediction across heterogeneous experimental conditions.

1 Introduction

Lipid nanoparticles (LNPs) have become a central platform for RNA-based therapeutics and vaccines, enabling efficient delivery of messenger RNA (mRNA) and small interfering RNA (siRNA) to target tissues. Among the components of LNPs, ionizable lipids (ILs) play a particularly critical role due to their pH-dependent molecular properties, which facilitate endosomal escape and cytoplasmic release of RNA cargos. Consequently, the discovery of effective ionizable lipids and the rational design of LNP formulations are essential for achieving robust in vivo RNA delivery.

Despite their importance, systematic experimental evaluation of ionizable lipid candidates remains a major bottleneck. In vivo screening is inherently expensive and time-consuming, as each candidate lipid requires formulation optimization and animal testing. As a result, only a limited subset of the vast chemical space of ionizable lipids can be experimentally explored. Artificial intelligence (AI)-based predictive models have therefore attracted increasing interest as a means to prioritize promising lipid candidates and reduce experimental burden by predicting RNA transfection potency directly from molecular structure.

However, the development of accurate AI-based models is severely constrained by the limited availability of in vivo data. While aggregating datasets from multiple experimental studies could potentially increase training data size, such integration is complicated by substantial heterogeneity in experimental conditions. Delivery efficiency is strongly influenced by factors such as cargo type, delivered gene, target organ, and route of administration, leading to condition-dependent measurement scales and biological contexts. In the absence of shared reference ionizable lipids or reference LNP formulations, conventional normalization strategies are often insufficient, and indiscriminate pooling of heterogeneous datasets risks

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introducing confounding effects.

In this work, we investigate the feasibility of condition-based integration of heterogeneous in vivo LNP delivery datasets. We hypothesize that although pooling data across fundamentally different experimental conditions is inappropriate, datasets sharing a minimal set of experimental parameters—cargo type, delivered gene, target organ, and administration route—can be meaningfully integrated even when LNP composition differs and no reference lipid is available. To address this question, we develop a condition-aware predictive modeling framework that leverages molecular representations of ionizable lipids and systematically evaluates performance under both random-split and condition-held-out settings. By doing so, we demonstrate that condition-based pooling enables strong within-condition prediction and candidate ranking, while also revealing clear limitations in cross-condition generalization. Together, these findings provide practical guidance for leveraging heterogeneous in vivo datasets in AI-driven LNP design under data-limited settings.

2 Results

2.1 Dataset characterization and experimental condition definition

We curated an in vivo LNP delivery dataset comprising 1,016 samples collected under diverse experimental settings [1]. Each sample was associated with a specific combination of cargo type (mRNA or siRNA), delivered gene, target organ, and route of administration, as well as corresponding ionizable lipid structures and LNP compositions. The delivery outcome was quantified using unnormalized delivery readouts, reflecting experimentally measured transfection potency without reference-based normalization.

To enable principled data integration, we defined an experimental condition as a unique combination of cargo type, delivered gene, target organ, and administration route. Under this definition, the dataset was partitioned into eight distinct experimental conditions, with sample sizes ranging from 22 to 260 (Table 1). Each condition exhibited markedly different target distributions in terms of mean, variance, and dynamic range. These differences underscore the challenge of directly comparing or normalizing delivery outcomes across conditions in the absence of reference LNPs.

Table 1: Summary of in vivo RNA delivery datasets grouped by experimental condition

Cargo type	Delivered gene	Target organ	Route	N
mRNA	hEPO	Muscle	IM	260
mRNA	FFL	Liver	IV	208
mRNA	FFL	Spleen	IV	176
mRNA	FFL	Muscle	IM	145
siRNA	FVII	Liver	IV	91
mRNA	Peptide barcode	Liver	IV	65
mRNA	FFL	Lung epithelium	IT	49
mRNA	FFL	Lung	IV	22
Total	-	-	-	1026

2.2 Baseline performance and impact of condition heterogeneity

As a baseline, we evaluated simple predictors that estimate delivery potency using global or condition-specific mean values. While condition-specific baselines reduced prediction error compared to a global mean, they failed to capture structure-dependent variations arising from differences in ionizable lipid chemistry and LNP composition. This observation highlights the need for models that can leverage molecular features while respecting condition-dependent measurement scales.

The pronounced heterogeneity in target distributions across conditions further emphasizes that naive aggregation or global normalization strategies are insufficient for integrating in vivo delivery data collected under diverse experimental settings.

2.3 Within-condition predictive performance

We first evaluated model performance using standard K-fold cross-validation, in which samples from the same experimental condition may appear in both training and test sets. Under this setting, the AI model achieved substantially improved performance relative to baseline predictors, with a mean root mean squared error (RMSE) of approximately 2.5 and a Spearman rank correlation of approximately. Importantly, strong rank correlation indicates that the model reliably captures relative differences in delivery potency among LNP formulations within the same experimental condition.

To further assess robustness to scale differences, we evaluated prediction accuracy using within-condition z-score normalization. The resulting zRMSE values remained close to unity, suggesting that the model effectively learns condition-specific structure–performance relationships rather than relying on absolute scale differences across conditions.

2.4 Cross-condition generalization analysis

To assess the model’s ability to generalize across experimental conditions, we performed group-based cross-validation and leave-one-experiment-out (LOEO) evaluation, in which entire experimental conditions were held out during training. Under these more stringent settings, predictive performance degraded substantially, with rank correlations approaching zero or becoming negative in several conditions. This result indicates that the learned relationships between ionizable lipid structure and delivery potency do not readily transfer across conditions with different cargo types, target organs, or administration routes.

Importantly, this degradation was observed even when experimental-condition embeddings were explicitly handled to avoid information leakage, confirming that the limitation reflects genuine biological and experimental differences rather than modeling artifacts.

3 Discussion

Our results provide important insights into the integration and utilization of heterogeneous in vivo LNP delivery datasets for AI-based prediction. The strong within-condition performance demonstrates that, when a minimal set of experimental parameters is held constant, datasets differing only in LNP composition can be effectively pooled and leveraged to learn meaningful structure–performance relationships. Crucially, this is achieved without reliance on reference ionizable lipids or reference LNP formulations, which are often unavailable in practice.

At the same time, the pronounced degradation in cross-condition generalization highlights a fundamental limitation of current modeling approaches. Differences in cargo type, delivered gene, target organ, and administration route introduce substantial biological and technical variability that cannot be readily normalized away or inferred solely from lipid structure. These findings underscore the importance of explicitly accounting for experimental context when developing predictive models for in vivo RNA delivery.

From a practical perspective, our study suggests a realistic and scalable strategy for expanding training datasets under data-limited conditions. Rather than attempting to normalize and pool all available in vivo data, we advocate a condition-based integration approach in which datasets sharing key experimental parameters are aggregated, enabling robust within-condition ranking and prioritization of ionizable lipid candidates. Such models are well-suited for guiding experimental design by narrowing the search space of candidate lipids within a given delivery context.

Future work may extend this framework by incorporating explicit representations of experimental conditions into the model, enabling improved generalization across conditions. Additionally, integrating in vitro data or mechanistic descriptors may further enhance predictive power and interpretability. Overall, our findings establish principled boundaries for data integration in AI-driven LNP design and provide a foundation for more efficient exploration of ionizable lipid chemical space in RNA therapeutics.

References

- [1] Witten, J., Raji, I., Manan, R.S. et al. (2025) Artificial intelligence-guided design of lipid nanoparticles for pulmonary gene therapy. *Nat Biotechnol* **43**, 1790–1799

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